Subha Maity

Assistant Professor

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Research Interests

Transfer learning, distribution shift, and algorithmic fairness.

Academic and Research Appointments

Oct 2024–present	**Assistant Professor** Department of Statistics and Actuarial Science, University of Waterloo, Waterloo, ON, Canada.
Summer 2022	**Research Intern** IBM Research, Cambridge, MA, USA.
Summers 2016, 2017	**Research Assistant** Johns Hopkins School of Public Health, Baltimore, MD, USA.

Research Grants and Funding

2026–2032	**NSERC Discovery Grant** CAD 185,000; additional supplement of CAD 12,500.
2024–2029	**University of Waterloo Startup Grant** Principal Investigator. CAD 60,000.

Education

2018–2024	**PhD in Statistics, University of Michigan** Ann Arbor, MI, USA. Advisors: Moulinath Banerjee and Yuekai Sun. Dissertation: *A study of transfer learning problems and their connections to algorithmic fairness*.
2016–2018	**Master of Statistics, Indian Statistical Institute** Kolkata, India. Honors thesis advisor: Anil K. Ghosh.
2013–2016	**Bachelor of Statistics, Indian Statistical Institute** Kolkata, India.

Publications and Manuscripts

* denotes equal contribution where documented; † denotes a supervised Waterloo trainee. Publications are listed in reverse chronological order within each category. Manuscripts under review are listed separately.

Journal Articles

J1. M. Cheng†, **S. Maity**, Q. Tian, and P. Li (2026). Transfer Learning under Group-Label Shift: A Semiparametric Exponential Tilting Approach. *Scandinavian Journal of Statistics, accepted*. [Preprint]

J2. S. Somerstep*, F. Maia Polo*, A. F. M. de Oliveira, P. Mangal, M. Silva, O. Bhardwaj, M. Yurochkin, and **S. Maity** (2026). CARROT: A Cost Aware Rate Optimal Router. *Annals of Applied Statistics, accepted*. [Preprint]

J3. S. Roy*, **S. Maity***, S. Xue*, M. Yurochkin, and Y. Sun (2025). How does overparametrization affect performance on minority groups? *Transactions on Machine Learning Research*. [Paper]

J4. **S. Maity**, D. Dutta, J. Terhorst, Y. Sun, and M. Banerjee (2024). A linear adjustment-based approach to posterior drift in transfer learning. *Biometrika 111(1), 31–50*. [Paper]

J5. **S. Maity**, Y. Sun, and M. Banerjee (2022). Minimax optimal approaches to the label shift problem in non-parametric settings. *Journal of Machine Learning Research 23(346), 1–45*. [Paper]

J6. **S. Maity**, Y. Sun, and M. Banerjee (2022). Meta-analysis of heterogeneous data: Integrative sparse regression in high dimensions. *Journal of Machine Learning Research 23(198), 1–50*. [Paper]

J7. R. Bhattacharyya, A. Burman, K. Singh, S. Banerjee, **S. Maity**, A. Auddy, S. K. Rout, S. Lahoti, R. Panda, and V. Baladandayuthapani (2022). Role of multiresolution vulnerability indices in COVID-19 spread in India: A Bayesian model-based analysis. *BMJ Open, 12(11):e056292*. [Paper]

Refereed Conference Papers

C1. D. Bracale*, **S. Maity***, Y. Sun, and M. Banerjee (2025). Learning the distribution map in reverse causal performative prediction. *AISTATS; Proceedings of Machine Learning Research, 258:973–981*. [Paper]

C2. S. Somerstep, Y. A. Ritov, M. Yurochkin, **S. Maity**, and Y. Sun (2025). Limitations of refinement methods for weak-to-strong generalization. *COLM 2025*. [Paper]

C3. D. Bracale*, F. Maia Polo*, **S. Maity***, S. Somerstep*, M. Banerjee, and Y. Sun (2025). Microfoundation inference for strategic prediction. *AISTATS; Proceedings of Machine Learning Research, 258:919–927*. [Paper]

C4. F. Maia Polo*, **S. Maity***, M. Yurochkin, M. Banerjee, and Y. Sun (2024). Weak supervision performance evaluation via partial identification. *Advances in Neural Information Processing Systems, 37 (NeurIPS)*. [Paper]

C5. **S. Maity**, M. Agarwal, M. Yurochkin, and Y. Sun (2024). An investigation of representation and allocation harms in contrastive learning. *ICLR 2024*. [Paper]

C6. L. Ngweta, M. Agarwal, **S. Maity**, A. Gittens, Y. Sun, and M. Yurochkin (2024). Aligners: Decoupling LLMs and alignment. *Findings of the Association for Computational Linguistics: EMNLP 2024, pp. 13785–13802*. [Paper]

C7. L. Ngweta*, **S. Maity***, A. Gittens, Y. Sun, and M. Yurochkin (2023). Simple disentanglement of style and content in visual representations. *ICML; Proceedings of Machine Learning Research, 202:26063–26086*. [Paper]

C8. **S. Maity**, M. Yurochkin, M. Banerjee, and Y. Sun (2023). Understanding new tasks through the lens of training data via exponential tilting. *International Conference on Learning Representations (ICLR)*. [Paper]

C9. **S. Maity***, D. Mukherjee*, M. Banerjee, and Y. Sun (2023). Predictor-corrector algorithms for stochastic optimization under gradual distribution shift. *International Conference on Learning Representations (ICLR)*. [Paper]

C10. **S. Maity***, S. Xue*, M. Yurochkin, and Y. Sun (2021). Statistical inference for individual fairness. *ICLR 2021*. [Paper]

C11. **S. Maity***, D. Mukherjee*, M. Yurochkin, and Y. Sun (2021). Does enforcing fairness mitigate biases caused by subpopulation shift? *NeurIPS 2021*. [Paper]

Short Papers

S1. S. Bakshi* and **S. Maity*** (2023). Bayes classifier cannot be learned from noisy responses with unknown noise rates. *ICLR Tiny Papers 2023*. [Paper]

S2. B. C. Kwon, U. Kartoun, S. Khurshid, M. Yurochkin, **S. Maity**, D. G. Brockman, A. V. Khera, P. T. Ellinor, S. A. Lubitz, and K. Ng (2022). RMExplorer: A visual analytics approach to explore the performance and fairness of disease risk models on population subgroups. *IEEE Visualization and Visual Analytics (VIS), short paper, pp. 50–54*. [Paper]

Preprints and Manuscripts

P1. **S. Maity** (2026). Efficient Inference for U-Statistic Targets under Subpopulation Shift. *Submitted to AISTATS 2027*.

P2. D. Bracale and **S. Maity** (2026). Accuracy Is Not Identification: A Local Minimax Theory for Pretrained Model Libraries. *Submitted to AISTATS 2027*.

P3. A. Chakraborty and **S. Maity** (2026). The Statistical Cost of Adaptation in Multi-Source Transfer Learning. *Submitted to JMLR*. [Preprint]

P4. M. Xu†, **S. Maity**, and J. Dubin (2026). Robust inference for risk heterogeneity under group imbalance. *To be submitted to Statistics in Biosciences*. [Preprint]

P5. M. Xu†, **S. Maity**, and J. Dubin (2025). Diagnosis-based mortality prediction for intensive care unit patients via transfer learning. *Under review at the Canadian Journal of Statistics*. [Preprint]

Research Supervision

Current Trainees

Sep 2025–present **Mengqi Xu – PhD student, University of Waterloo**
Joint supervision with Joel Dubin.
Project: *Some advances in transfer learning methods for intensive-care unit datasets.*

Current **Nora Luchun – MMath research trainee, University of Waterloo**

Past Trainees

Past trainee **Taoyue Chen – MMath research trainee, University of Waterloo**

Postdoctoral Supervision

Oct 2024–Dec 2026 **Manli Cheng – Postdoctoral Fellow, University of Waterloo**
Joint supervision with Qinglong Tian and Pengfei Li.

Teaching Experience

Course Instructor – University of Waterloo

Fall 2026 **STAT 845 – Statistical Concepts for Data Science**

Fall 2025 **STAT 441/841 and CM 763 – Statistical Learning: Classification**

Fall 2025 **STAT 845 – Statistical Concepts for Data Science**

Spring 2025 **STAT 333 – Stochastic Processes I**

Winter 2025 **STAT 440/840 and CM 761 – Computational Inference**

Graduate Teaching – University of Michigan

Fall 2018 **STAT 250 – Introduction to Statistics**
Lab instructor and teaching assistant.

Honors and Awards

2024 **Rackham Dissertation Award**
University of Michigan.

2024 **Outstanding Dissertation Award**
Department of Statistics, University of Michigan.

2023 **Predoctoral Fellowship**
Rackham Graduate School, University of Michigan.

2022 **Future Leader**
Michigan Institute for Data Science, University of Michigan.

2018 **Outstanding First-Year PhD Student**
Department of Statistics, University of Michigan.

Talks and Presentations

Invited Talks

2026 **The Statistical Cost of Adaptation in Multi-Source Transfer Learning**
International Symposium on Nonparametric Statistics (ISNPS), Thessaloniki, Greece.

2024 **Learning the distribution map in reverse causal performative prediction**
CFE-CMStatistics, London, United Kingdom.

2024 **Tackling posterior drift via linear adjustments and exponential tilts**
Department seminars: University of Waterloo, Waterloo, ON, Canada; University of Texas at Dallas, Dallas, TX, USA; University of South Florida, Tampa, FL, USA.

2024 **Investigations of algorithmic biases caused by underrepresentation**
Department seminar, University of California San Diego, San Diego, CA, USA.

2023 **Understanding new tasks through the lens of training data via exponential tilting**
INFORMS Annual Meeting, Phoenix, AZ; ATD Workshop, Fairfax, VA.

2022 **Statistical perspectives on federated learning**
INFORMS Annual Meeting, Indianapolis, IN.

Contributed Talks and Posters

2026 **The Statistical Cost of Adaptation in Multi-Source Transfer Learning**
Statistical Society of Canada Annual Meeting, Hamilton, ON, Canada.

2025 **CARROT: A Cost Aware Rate Optimal Router**
Statistical Society of Canada Annual Meeting, Saskatoon, SK, Canada.

2023 **Understanding new tasks through the lens of training data via exponential tilting**
Michael Woodroofe Memorial Conference, Ann Arbor, MI; Purdue Data Science Workshop, West Lafayette, IN.

2022 **Does enforcing fairness mitigate biases caused by subpopulation shift?**
INFORMS Annual Meeting, Indianapolis, IN; Future Leaders Summit, Ann Arbor, MI; Michigan Student Symposium for Interdisciplinary Statistical Sciences, Ann Arbor, MI.

2021 **Statistical inference for individual fairness**
International Conference on Learning Representations, virtual.

2021 **Does enforcing fairness mitigate biases caused by subpopulation shift?**
Neural Information Processing Systems, virtual.

2021 **Minimax optimal approaches to the label shift problem in non-parametric settings**
Michigan Student Symposium for Interdisciplinary Statistical Sciences, Ann Arbor, MI.

2020 **Minimax optimal approaches to the label shift problem in non-parametric settings**
Joint Statistical Meetings, virtual.

Professional Service

Academic Service

Oct 2024–present **Seminar Committee**
University of Waterloo.

Fall 2021; Winter 2022 **Co-Chair – Student Seminar Committee**
University of Michigan.

Peer Review

Annals of Statistics; *Journal of the American Statistical Association*; *Journal of Computational and Graphical Statistics*; *Journal of Statistical Planning and Inference*; *Bernoulli*; *Statistical Analysis and Data Mining*; *Statistical Science*; Neural Information Processing Systems (NeurIPS).